

Experimental Implementation and Hardware

All experiments were executed using Python code on the project's compute server. The final manuscript should report the exact CPU, GPU, memory, operating environment, container or virtual-environment configuration, and relevant software versions from the recorded runtime provenance. These hardware details are not established by the current draft and should not be inferred.

The implementation was divided into two principal repositories. The first was the public **STraTS** repository, which was not originally authored by us. We modified and extended that code for the CliniCause datasets, preprocessing and identifier contracts, proxy-prediction targets, artifact and provenance formats, and integration with the complete workflow. The second was the **causal-analysis pipeline**, which we developed from scratch. This pipeline constructs the analysis inputs, applies the dataset-specific adjustment specifications, invokes the causal estimators, performs matching, aggregates the estimates, and produces the reported artifacts.

The causal estimators themselves were integrated from established external implementations rather than reimplemented. **CausalForestDML** and **LinearDML** were used through the open-source **EconML** Python package. EconML implements double/debiased machine-learning estimators using cross-fitting and orthogonalized nuisance estimation. LinearDML uses a linear final-stage effect model, whereas CausalForestDML uses a causal-forest final stage to estimate heterogeneous treatment effects. Our code supplied the exposure, outcome, adjustment variables, effect-modification variables, model configuration, seeds, and output handling around these estimators.

CausalPFN was integrated from the implementation and pretrained-model interface released in the authors' public CausalPFN repository. We used that implementation within our pipeline rather than reproducing its architecture or pretraining procedure. The matching-based analysis, by contrast, was implemented by Kobi specifically for this project.

Accordingly, the surrounding CliniCause causal pipeline - including data preparation, variable-role assignment, estimator invocation, matching, result collection, provenance, and integration with the constructed resources - was our own implementation. STraTS and the estimator implementations were adapted or integrated from existing open-source projects, while the CliniCause causal pipeline and matching implementation were developed for this work.

Hardware reporting note. Replace the first paragraph's general wording with the exact server specification only after extracting it from the validated runtime records. Do not reconstruct hardware or environment details from memory.